

# SRI HARSHA

System Architecture Architect | AI/Agentic Strategist | Telecom Systems

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## PROFESSIONAL SUMMARY

**22+ years building telecom and AI systems** — from kernel-level packet processing to agentic AI platforms. Architect who has taken **3 products from R&D to commercial deployment**: world's first commercial 3GPP Femto Gateway (SFR France), world's first 5G-NSA deployment (T-Mobile), and supporting Reliance Jio's greenfield all-IP LTE network.

Unique career arc spanning every layer of the stack: bare-metal C/C++ with DPDK, cloud-native Kubernetes microservices in Golang, and now **full-stack AI/LLM/Agentic infrastructure**. Pioneered CI/CD, observability platforms, and test automation *before* industry standardization (pre-Jenkins, pre-Prometheus era).

Currently driving strategic AI adoption at Ribbon Communications: built on-premises LLM clusters (12 GPUs, 230GB VRAM), deployed RAG chatbots saving \$100K+/year, engineered agentic RFX systems reducing sales response from months to days, and championing **agentic software dark factories** — where 16+ years of architecture experience writing PRDs, FSDs, DSDs, and revision design specifications directly fuel autonomous code generation at 100x velocity.

## KEY HIGHLIGHTS

| 22+ Years        | 3 Products to Market | 3B Events/Hour       | US Patent        | 1:2 GPUs / 230GB VRAM |
|------------------|----------------------|----------------------|------------------|-----------------------|
| Total Experience | R&D to Commercial    | Real-time Processing | QoS Optimization | On-Prem AI Cluster    |

## WORK EXPERIENCE

### Ribbon Communications — Research Labs | System Architecture Architect | 2021 – Present

- Ribbon Analytics Platform**: Core architect on cloud-native, real-time big-data analytics platform — sensor network collecting CDRs, performance metrics, SIP signaling PDUs, faults, logs, and alarms. CDR-to-SIP correlation for end-to-end visibility. Architected systems processing **3 billion events/hour**, serving 2.5M customers at Airtel India; designed to scale to 400 billion events/hour.
- 5G VoNR Analytics**: Enhanced Ribbon Analytics for full 5G stack (AMF, AUSF, SMF, gNodeB, UPF, PCF, UDM, CHF, NRF, NSSF, SCP); quasi-NWDAF; first VoNR deployment at Airtel India.
- Acumen Platform**: Key contributor to Ribbon's Acumen — AIOps and automation platform for autonomous networking, with out-of-the-box apps and Acumen Builder for low-code/no-code custom app creation.
- Acumen AI Cluster**: Built 5-node K8s cluster with 12 GPUs (8x L40S + 4x Tesla M60), 230 GB VRAM, 192 virtual GPU slices — a scaled-down Amazon Bedrock equivalent.
- Acumen AI Platform**: Self-service platform (FastAPI + Celery + Flask + MCP Servers) on distributed Kubernetes infrastructure for enterprise AI workloads.
- RAG Chatbot**: Pioneered on-premises LLM deployment (NVIDIA A100/H100/L40S); 100+ GB data, 10M+ tokens served, **\$100K+ annual savings** for 1,000s of users with full data sovereignty.
- RFX Responder**: Agentic AI system addressing 15,000 technical questions in minutes; reduced sales first response from months to days; accuracy improved 75% to 87%.
- LLM Fine-tuning**: Built pipelines using Unsloth/LoRA/QLoRA/RLHF; created custom LLMs and SLMs for product-specific knowledge; deployed Langflow with ReAct patterns, memory, and context engineering.
- LLM Observability & Evaluation**: Extensively used Arize-Phoenix and Langfuse for LLM tracing, evaluation, dataset creation, eval set curation, and systematic validation of various LLMs.
- Sentry Integration**: Deployed Sentry for production issue tracking — monitoring exceptions, performance bottlenecks, and release health.
- MINT Project**: Mobile Inband Telemetry using Intel Tofino switch with P4 programming — processing Terabits/second; metadata creation for ML-based network analysis.
- MWC Demos (2022–Present)**: Prepared and delivered customer-facing POC demos at Mobile World Congress — AI-powered analytics, agentic AI platforms, and futuristic network intelligence.
- CXO AI Strategy Advisory**: Trusted advisor to VPs, SVPs, PMs, and CXOs on AI strategy, roadmap preparation, executive slides, and technology investment decisions.
- AI/ML Steering Committee**: Key member driving enterprise GenAI adoption; introduced Codeium, Windsurf, Cursor across organization.
- Market Strategy**: Competitive intelligence, product viability assessments, MVP definition; prototyped next-gen analytics using open-source free5GC.

### Cisco Systems India Pvt Ltd | Software Engineer IV — Mobility Unit | 2015 – 2021

- **5G-NSA (T-Mobile):** Architected 5G-NSA solution on MME/PGW for **world's first 5G-NSA deployment**; implemented path update procedures and extended QoS attributes.
- **5G SMF (Cloud-Native):** Owned QoS Modeling — heart of Cisco's industry-first Cloud Native Standalone 5G Core; managed SMF-UPF and SMF-CHF interfaces.
- **MPTCP Prototype:** Prototyped Multipath TCP on DPDK for high-throughput networking; ported TCP stack to DPDK; explored scheduling algorithms (Reno, Vegas, Lowest RTT).
- **SaMOG WiFi Gateway:** Designed S2a WiFi Offload features (Trusted WiFi in LTE, roaming, web auth); deployed across Indian colleges and public WiFi.
- **Call Flow Diagram Tool:** Built sequence diagram generation from distributed K8s pods — single flow view across pods.
- **IEC-5 Startup — 5G-SPACE:** Semifinalist among 400+ submissions (top 20) in Cisco CTO organization.

### Samsung Research Institute, Bangalore | Chief Engineer | 2014 – 2015

- **Reliance Jio:** Led TAC team supporting **world's first and largest all-IP 4G/LTE network** — 125K gNodeBs with Evolved Packet Core.
- **EPC Operations:** OAM of MME, SGW, PGW; managed KPIs for 100,000+ eNodeBs; managed team of 7 engineers.

### Nokia Solutions & Networks | Product Architect / OAM Expert | 2008 – 2014

- **World's First Femto Gateway:** Architected and delivered **world's first commercial 3GPP Femto Gateway (HNBGW)** for SFR, France — lab to field deployment.
- **FCAPS @ Scale:** 60,000 concurrent Femto APs with **99.999% availability** using active/backup redundancy.
- **Agile Transformation:** Led transformation of 120-member cross-functional team; Scrum Master for multiple teams.
- **CI/CD Pioneer:** Handcrafted CI/CD system using Perl and Expect **before Jenkins existed**; automated testing framework with 5,000+ test cases.
- **Health Monitor Console:** Observability platform predating Prometheus/Grafana — BU-level innovation finalist.
- **IMS CSCF Redesign:** Converted bare-metal IMS CSCF to cloud-native using DPDK; achieved 400% SIP packet processing efficiency.
- **Features:** Multiple-CN, lUflex, AGPS, Location Locking, Area-based Charging, RADIUS/EAP-SIM (RFC2865/RFC4186), SOAP Web Service, MySQL scaling to 50M entries, flow-based logging, NETCONF/YANG, SNMP, TR-069/TR-196.

### Motorola India Pvt Ltd | Software Engineer | 2004 – 2008

- **3G Flex & Feature Database:** Core developer for P2K phone platform — NV memory and file system management for 3G handsets delivered across 5 continents.
- **Scaling:** Scaled from few to **172 simultaneous customers** at Razr peak — 6X customer scaling with 4X delivery performance.
- **OTA Updates:** Architected Over-The-Air update system for 3G phones; built E2E Validation Tool (VC++) and Advanced Flex Tool.

## TECHNICAL SKILLS

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**AI / ML / Agentic:** LLMs, SLMs, RAG, LoRA, QLoRA, RLHF, Unsloth, vLLM, Ollama, LangChain, LangGraph, LlamaIndex, Langflow, Ragflow, Dify, HuggingFace, PyTorch, OpenAI API, Claude Code, GitHub Copilot, Windsurf, Cursor, Codeium, Fine-tuning Pipelines, Document Classification, Sentiment Analysis, Prompt Engineering, Custom LLM/SLM Creation, TTS/STT Models, MLOps, Synthetic Data Pipelines, MCP Servers, Agentic Workflows, ReAct Patterns, Agent Memory, Context Engineering, LLM Selection Strategy, Embedding Model Selection

**Telecom & Networking:** 5G Core (SMF, AMF, UPF, PCF, AUSF, UDM, CHF, NRF, NSSF, SCP), 4G/LTE (MME, SGW, PGW), 3G/UMTS, VoLTE, VoNR, IMS (HSS, CSCF, DR), Femto Gateway (HNBGW), SaMOG, EPC, NWDAF, FCAPS, OAM, 3GPP Specifications, IETF Standards, free5GC

**Programming Languages:** C/C++ (20+ yr), Python (10+ yr), Golang (8+ yr), Perl (15+ yr), Shell Scripting, JavaScript, PHP, VC++

**Protocols (40+):** IP/IPv4/IPv6, TCP/UDP, GTPv2, RADIUS, Diameter, EAP-SIM, EAP-AKA, IPSec/IKEv2, gRPC/Protobuf, SNMP, NETCONF/YANG, TR-069, SIP, MPTCP, HNBAP, RANAP, GRE/EoGRE/PMIP, SOAP, REST

**Cloud & Infrastructure:** Kubernetes, Docker, Cloud-Native Microservices, AWS Bedrock, GCP, Azure, Distributed Systems, HA (99.999%), Containerization

**Systems & Performance:** DPDK, P4 Programming (Intel Tofino), eBPF, Multi-core Programming, Linux Internals, Real-time Packet Processing, Terabits/sec Processing

**Data & Databases:** ClickHouse, Hadoop, Parquet, Milvus (Vector DB), Redis, PostgreSQL, MySQL, Chroma

**Observability:** Prometheus, Grafana, Arize-Phoenix (LLM Evals/Tracing/Dataset Creation), Langfuse (LLM Observability/Eval Sets/Prompt Tracking), Langsmith, Sentry, OpenTracing

**DevOps & Tools:** Git, SVN, ClearCase, Jenkins, CI/CD, JIRA, Confluence, GCC/G++, GDB, Valgrind, Wireshark, strongSwan

## STRATEGY & ADVISORY

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Trusted AI strategy advisor to CXOs, SVPs, VPs, and Product Managers — translating complex GenAI/LLM/Agentic concepts into actionable business strategies, build-vs-buy decisions, and investment prioritization.

- **MWC Customer-Facing POC Demos (2022–Present):** Prepared and delivered multiple customer-appealing POC demos at Mobile World Congress — AI-powered analytics, agentic AI platforms, and futuristic network intelligence showcases.
- **CXO AI Strategy Advisory:** Regularly consulted by CXOs, SVPs, and VPs on AI strategy — helped leadership understand build vs. buy decisions, open-source vs. proprietary AI tradeoffs, and investment prioritization.

- **Executive Presentations:** Prepared AI strategy decks, technology roadmap slides, and competitive analysis for senior leadership — enabling confident communication of AI vision to customers, partners, and investors.
- **Product Manager AI Enablement:** Guided PMs on integrating AI capabilities into product roadmaps — identifying which features benefit from LLMs, RAG, fine-tuning, or agentic workflows.
- **AI/ML Steering Committee:** Core member driving company-wide GenAI adoption; introduced and evaluated developer tools; moderated internal AI/ML forum.

## 100X AGENTIC MULTIPLIER — WHY ARCHITECTURE EXPERIENCE ENABLES DARK FACTORIES

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With 16+ years of writing PRDs, FSDs, DSDs, and revision design specifications across Nokia, Samsung, Cisco, and Ribbon, I produce specifications with the precision AI agents need. Combined with hands-on agentic infrastructure building, this enables autonomous code generation at 100x velocity.

- **Specification Precision:** 16+ years of 3GPP-compliant specs eliminates ambiguity — agents generate production-quality code on first pass.
- **Full-Stack Context:** From DPDK kernel-level to Kubernetes to LLM fine-tuning — complete architectural context for agents.
- **Agentic Infrastructure Builder:** Built Acumen AI Platform (FastAPI + Celery + MCP Servers) + Langflow with ReAct, memory, and context engineering.
- **LLM Fine-tuning:** Fine-tuned LLMs on product-specific domain knowledge (LoRA/QLoRA/RLHF). Built RAG systems over 100GB+ documentation.
- **LLM & Embedding Selection:** Deep expertise choosing the right LLM for each task (thinking, summarizing, categorizing, code generation) and right embedding model for each data type.
- **Agentic Tool Proficiency:** Claude Code, GitHub Copilot, Windsurf, Cursor, Langflow, LlamaIndex (pre-LlamaCloud), LangChain/LangGraph (certified), MCP Servers.
- **Pioneer Mentality:** CI/CD before Jenkins, observability before Prometheus, dashboards before Grafana — now building agentic software factories before mainstream.

## AI LAB & EXPERIMENTATION COOKBOOK

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Active personal AI experimentation lab — a living cookbook of hands-on projects exploring cutting-edge AI/ML. GitHub: [github.com/reachsriharsha/mycookbook](https://github.com/reachsriharsha/mycookbook)

- **Voice AI:** Built real-time voice agent using LiveKit + Sarvam Saarikaa STT v2.5 + Ollama Gemma3:4B + Sarvam Bulbul TTS v2. Tested multiple speakers for Indian English voice quality. Gemma video/audio fine-tuning.
- **LLM Fine-tuning Lab:** Comprehensive Unsloth cookbook — LoRA/QLoRA on Llama 3.2 3B, PDF-to-QA pipelines (PyMuPDF), GGUF export for Ollama, ROUGE/BERTScore evaluation. Classification fine-tuning, Llamini experiments.
- **RAG & Compliance AI:** RFQ compliance assessment with DeepSeek-R1:8B + LangChain + Chroma (multi-stage retrieval: cosine similarity, cross-encoder re-ranking, fuzzy matching). ExcelRAGProcessor for intelligent table chunking. Milvus backup/restore tooling.
- **Agentic Frameworks:** Custom Langflow components (KB List Fetcher, Ollama integration, regex processors). Go-to-Celery bridge. 460+ section Self-Service AI Platform architectural wiki.
- **Full-Stack:** Firebase + FastAPI RBAC authentication. OTP auth with token refresh. Data Explorer (Flask). LangChain dynamic prompting techniques.
- **NLP & Deep Learning:** LSTM networks, text summarization, POS tagging, NLTK, HuggingFace text-to-image pipelines.

## PRODUCT & LEADERSHIP

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Product Lifecycle (Concept to Market), PRD/FSD/DSD Authoring, Revision Design Specifications, MVP Definition, Market Analysis & Competitive Intelligence, Agile Transformation (120 members), Scrum Master, Cross-functional Team Leadership, TAC Management, Mentoring Engineers, Customer Interface & Field Deployment.

## CERTIFICATIONS

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**Introduction to LangGraph** — LangChain Academy | December 2025

**Foundation: Introduction to LangChain — Python** — LangChain Academy | December 2025

**Agentic AI!** — DeepLearning.AI | October 2025

**Building AI Voice Agents for Production** — DeepLearning.AI | September 2025

**Advanced NLP Techniques: LoRA for Fine-Tuning Llama3 LLMs** — Udemy | February 2025

**Machine Learning for Absolute Beginners — Level 1** — Udemy | February 2023

## PATENT & AWARDS

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**US Patent US011350319B2:** QoS optimization for inter-RAT handover

- IEC-5 Semifinalist — 5G-SPACE venture, top 20 of 400+ at Cisco

- BU Innovation Finalist (x2) — Real Time Analyzer for Logs & Health Monitor Console at Nokia
- Most Inspiring Person — Femto Gateway team at Nokia
- Spot Awards — CI framework development & FGW feature integration
- Star Scrum Team Award — Nokia

## EDUCATION

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**M.S. Software Systems** — BITS Pilani (Distance)

**B.E. Computer Science & Engineering** — AIT, Chickmagaluru (VTU)

## PERSONAL

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Date of Birth: March 28, 1983 | Languages: Kannada, English, Hindi | Interests: Technology exploration, photography, traveling, reading, music